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# Topographic accessibility and proto-urban settlement organization in Neolithic China

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Cities emerged from settlement systems that predated formal roads, mature states and institutionalized urban governance, but the large-scale spatial organization of these proto-urban systems is still poorly understood. Here we examine how terrain-derived accessibility related to settlement organization across Neolithic China between 5000 and 2000 BCE, combining a newly compiled georeferenced archeological dataset with a measure of overland accessibility derived from topography. Settlement intensity became increasingly concentrated along accessible corridors between 5000 and 2500 BCE. Between 2500 and 2000 BCE, this association weakened sharply, although spatial clustering remained strong and the principal concentration of settlements shifted toward intermediate-accessibility settings. These findings show that terrain accessibility structured proto-urban settlement organization before mature states and formal transport infrastructure, but that this relationship was historically contingent rather than fixed. They provide a long-run East Asian perspective on how the spatial foundations of urban development can be reorganized as settlement systems grow more complex.

The emergence of cities was a major transformation in human social organization[1], but cities did not appear fully formed. They developed from earlier settlement systems that predated formal roads, mature states and institutionalized urban governance. Understanding how these systems were organized is, therefore, essential to explain the spatial foundations of urban development. Physical geography has long been associated with the distribution of cities and economic activity—from Bronze Age Anatolian cities along natural overland routes to maritime connectivity in the Iron Age Mediterranean, Atlantic trade in early modern Europe and the contemporary global economy[2–5]. Yet, whether landscape accessibility structured settlement systems before state formation remains poorly understood.

Here "settlement" refers to archeologically recorded habitation sites and their aggregate spatial distribution. We use "proto-urban settlement organization" to describe the system-level arrangement of these sites before the emergence of larger, more differentiated and institutionally organized cities. The term does not imply that individual sites were urban. Accordingly, our site-count measure captures settlement presence and intensity, rather than city size, population, functional differentiation or governance. We ask whether accessibility afforded by the natural landscape was systematically associated with the organization of these settlement systems, and whether that relationship changed as they grew in scale and complexity.

Neolithic China provides an unusually informative setting in which to address this question. Its broad cultural sequence and archeological development from the Neolithic to early state formation have been extensively documented[6–8]. During the later Neolithic, this trajectory included the emergence of major centers whose scale and differentiation have been interpreted as part of the transition toward early urbanism and the rise of Erlitou[9]. Long-term regional surveys,

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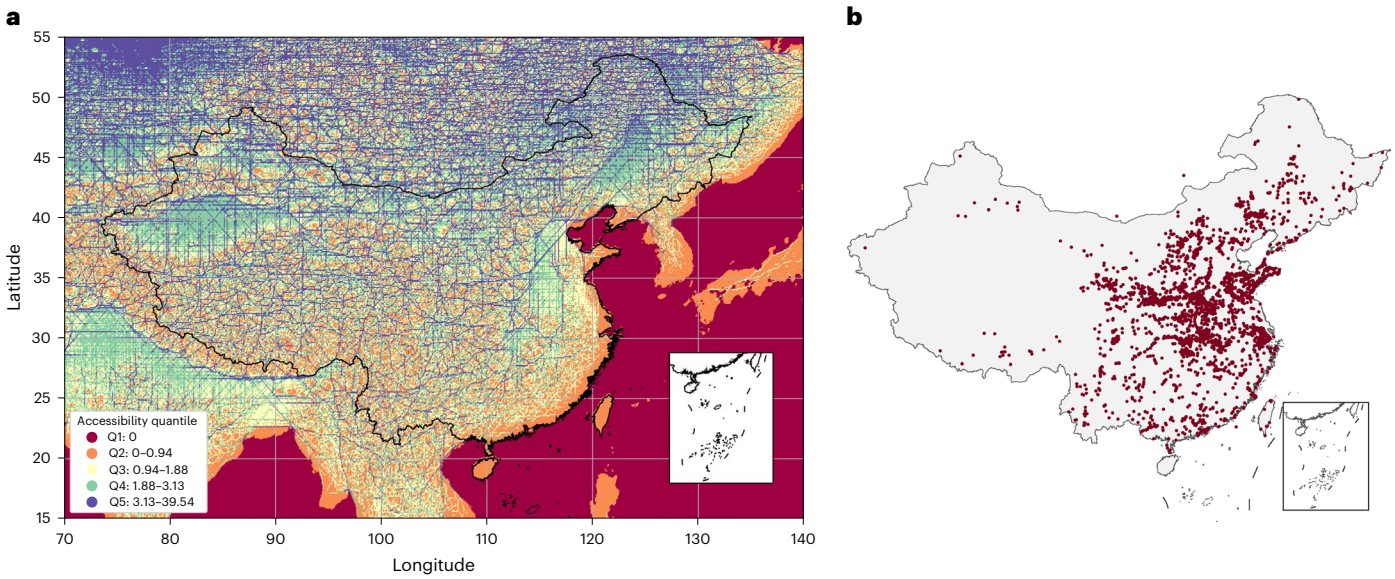



**Fig. 1 | Terrain accessibility and archeological site distribution. a**, Spatial distribution of the primary independent variable: the terrain-accessibility index computed using the NRS algorithm with landform constraints. The map reveals distinct terrain-defined corridors emerging from the underlying topographic morphology. **b**, Geographical distribution of the total archeological site sample ($N = 3,587$) used in this study. The dataset comprises Neolithic settlements synthesized from published archeological reports and systematic regional surveys. Basemaps cited from Standard Map Service, Ministry of Natural Resources of the People's Republic of China (no. GS(2024)0650; https://bzdt.ch.mnr.gov.cn/).

rescue excavations and archeological investigations prompted by modern construction and infrastructure development have produced a large body of georeferenced site records spanning the major Neolithic cultural phases between approximately 5000 and 2000 BCE. These records make it possible to examine settlement presence and intensity at a national scale and to trace how their spatial organization changed through time.

To examine this question, we combine the georeferenced archeological dataset with a terrain-derived measure of overland accessibility (Methods). The index uses elevation-derived walking-time friction and the geometry of macro-topographic corridors to summarize the relative ease of cross-regional terrestrial movement implied by terrain alone. We evaluate the relationship between this measure and settlement intensity using grid-based panel regressions and complementary spatial diagnostics, including global Moran's $I$, local indicators of spatial association (LISA), nearest-neighbor distance and Ripley's $L$. Because the index is constructed from terrain alone, we interpret it as a geometric property of the landscape rather than as a reconstruction of specific prehistoric routes, exchange flows, or riverine and maritime mobility.

The results indicate that the relationship between terrain-derived accessibility and settlement intensity was historically contingent rather than fixed. From the Early Neolithic (5000–4000 BCE) through the Middle Neolithic (4000–3000 BCE) and Late Neolithic I (3000–2500 BCE; Early Longshan), settlement intensity became increasingly concentrated along terrain-defined accessible corridors, with the estimated positive association remaining statistically significant across all three phases. In Late Neolithic II (2500–2000 BCE; Late Longshan), however, this long-running association weakened markedly and was no longer statistically distinguishable from zero in the principal specifications. Spatial clustering, nevertheless, remained pronounced: its principal concentration shifted away from the highest-accessibility corridors toward intermediate-accessibility settings, the contrast between concentrated cores and extensive low-density areas became sharper, and localized clusters persisted in topographically constrained, low-accessibility landscapes. Supplementary regional analyses further indicate that this phase-specific pattern was not spatially uniform and was most clearly expressed in the northern core of the core agricultural region. The Late Neolithic II change is, therefore, better understood as a reorganization of settlement aggregation than as simple spatial dispersal.

The timing of this reorganization broadly coincides with the widely discussed interval of hydroclimatic instability around 4.2-ka BP (refs. 10,11), although substantial debate remains over the timing, magnitude and regional consequences of this episode in China and elsewhere[12,13]. We do not directly test climatic mechanisms and treat this temporal correspondence only as one possible contextual clue, alongside settlement nucleation, regional differentiation, changes in mobility and land use, and social or ritual reorganization. These alternative interpretations are considered in the "Discussion" section. More broadly, the analysis provides a long-run perspective on how the spatial foundations of proto-urban settlement systems could be organized—and reorganized—before mature states and formal transport infrastructure.

## Results

### Spatial correspondence

We first examine the spatial correspondence between terrain accessibility and recorded settlement intensity. Figure 1a presents the terrain-accessibility surface (Methods), which summarizes the relative ease of movement implied by macro-topographic constraints at the study's grid resolution. Shaped by regional geomorphological structure, this surface reveals relatively persistent low-resistance transit zones and macro-corridor structure constrained by terrain. Figure 1b displays the distribution of archeological sites across the full sample, illustrating that the recorded archeological record is highly uneven spatially, forming prominent agglomerations in several macroscopic belt-like regions.

The compiled record expands markedly through time: total site counts increase from 1,604 in the Early Neolithic to 2,452 in Late Neolithic II for All China, with a parallel rise from 1,347 to 2,131 in the core agricultural region (Fig. 2a). The mean recorded settlement intensity increases across all terrain-accessibility quartiles, but its distribution remains uneven (Fig. 2b). A substantial share of recorded sites also lies outside the highest-accessibility quartile, anticipating the localized settlement pockets examined below (Fig. 2c).

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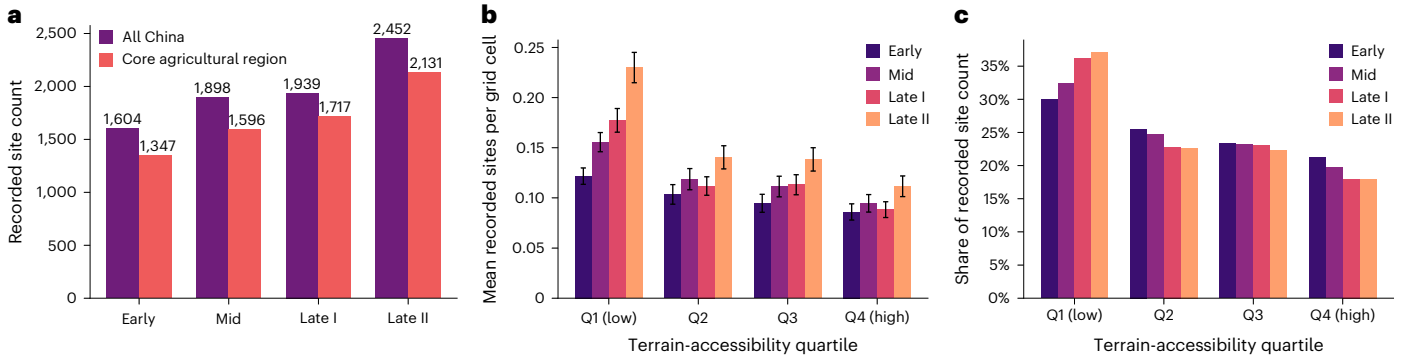



**Fig. 2 | Sample coverage and recorded settlement intensity across terrain-accessibility strata. a**, Recorded site totals for All China (Early, 1,604; Mid, 1,898; Late I, 1,939; Late II, 2,452) and the core agricultural region (1,347, 1,596, 1,717 and 2,131, respectively). **b**, Mean recorded site count per 0.25° × 0.25° grid cell across quartiles of terrain accessibility (Q1–Q4). Bar heights show arithmetic means and error bars show ±s.e.m. Each phase contains $n = 15{,}810$ grid cells: Q1, $n = 3{,}953$; Q2, $n = 3{,}952$; Q3, $n = 3{,}952$; Q4, $n = 3{,}953$. **c**, Percentage of the total recorded site count in each phase contributed by each accessibility quartile. No inferential tests were applied in this figure.

We next map the recorded settlement intensity across the four phases (Fig. 3a–d). Rather than a wholesale relocation of settlement, the maps primarily show an intensification of existing cores. The corresponding bivariate maps (Fig. 3e–h) show the intersection of settlement intensity and terrain accessibility, revealing both corridor-aligned agglomerations and localized clusters outside the most accessible landscapes.

## Baseline estimation

To estimate the association between recorded settlement intensity (proxied by site counts) and terrain accessibility, we use a baseline panel regression framework (Methods provides detailed specifications). In the analysis, we use the number of archeological sites within each grid cell as the dependent variable. After controlling for geographic variables (elevation, slope, caloric productivity and coastal proximity) and fixed effects, Table 1 reports the baseline regression results. Because settlement traditions varied substantially across Neolithic China, the regression is estimated first at the national level; its spatial uniformity is then assessed through regional sub-sample splits (Supplementary Note 4).

In the panel estimation covering the entire Neolithic period (Table 1, column 1), terrain accessibility exhibits a positive association with the recorded settlement intensity. For the full sample covering All China, the estimated coefficient on the terrain-accessibility index is 0.136 (two-sided $P = 0.014$). When restricting the sample to the core agricultural region, the coefficient is 0.426 (two-sided $P = 0.020$). Because the index is normalized to range between 0 and 1, these coefficients capture the expected association with moving from the least to the most accessible terrain at the macroscale defined by topographic geometry. For instance, in the core agricultural region, a full-range increase in this index (from 0 to 1) is associated with an increase of approximately 0.426 recorded sites per grid cell, conditional on the model controls and fixed effects.

## Temporal dynamics of the accessibility–settlement relationship

To examine how the relationship between terrain accessibility and recorded settlement intensity evolved over time, Table 1 (columns 2–5) reports the regression estimates disaggregated by Neolithic phase; the same coefficients and their 95% confidence intervals are shown in Fig. 4a. These estimates reveal a marked divergence in the temporal dynamics of recorded settlement distribution.

In the early and middle phases, the statistical association between terrain accessibility and recorded settlement intensity shows a steady and statistically significant increase. In the All China sample, the coefficient increases from 0.133 in the Early Neolithic to 0.167 in the Mid Neolithic, remaining relatively stable at 0.163 in the Late Neolithic I (all statistically significant). A similar, yet more pronounced, pattern emerges in the core agricultural region sample, where the coefficients rise from 0.413 to 0.470 and 0.462 across the same periods.

Yet, this longer-run pattern clearly weakens at the very end of the Neolithic (Late Neolithic II). The association between terrain accessibility and recorded settlement distribution weakens substantially, losing its significance across most specifications. In the core agricultural region sample, the coefficient drops from 0.462 in Late Neolithic I to 0.358 in Late Neolithic II. Similarly, in the All China sample, the coefficient falls from 0.163 to 0.083. The magnitude of this attenuation indicates a clear departure from the stronger associative patterns observed in earlier phases.

Supplementary analyses show that this phase-specific pattern is robust to alternative chronological partitioning, regional sample restrictions, alternative coastal-distance thresholds and the inclusion of a historical river-network proxy (Supplementary Notes 3–5). Re-estimation using a higher-resolution six-phase chronology yields the same broad result, and regional splits indicate that the relationship is not spatially uniform, with its clearest expression in the northern core. The accessibility coefficient also remains stable in direction and broad magnitude after adding distance to major historical rivers as an additional control (Supplementary Note 5), indicating that the phase-specific pattern is not simply an artifact of grid cells lying close to major rivers. An orthogonalized re-estimation—removing the 4.3% of index variance linearly attributable to river/coast proximity (variance inflation factor = 1.23)—confirms that the phase-specific pattern is fully preserved in the residual terrain-corridor signal (Supplementary Note 5). Site-area data are unevenly reported across the source archeological literature and we, therefore, use them only as a conservative sensitivity check rather than as a replacement outcome (Supplementary Note 2). Within the limits of these data, area-weighted specifications preserve the phase-specific pattern observed in the count-based baseline: positive in the earlier phases and clearly weaker in Late Neolithic II. The late weakening is unlikely to be explained solely by the equal-weight treatment of large centers and small hamlets.

## Spatial point-pattern analysis of settlement clustering

To probe the weakened association observed in the Late Neolithic II regressions, we conducted complementary spatial analyses using global Moran's $I$, LISA, nearest-neighbor distance and Ripley's $L$ function, with the point-pattern summaries stratified by absolute quartiles of terrain accessibility (Q1–Q4; Fig. 4b–f and Table 2). These diagnostics make spatial agglomeration directly observable rather than inferred solely from regression coefficients.

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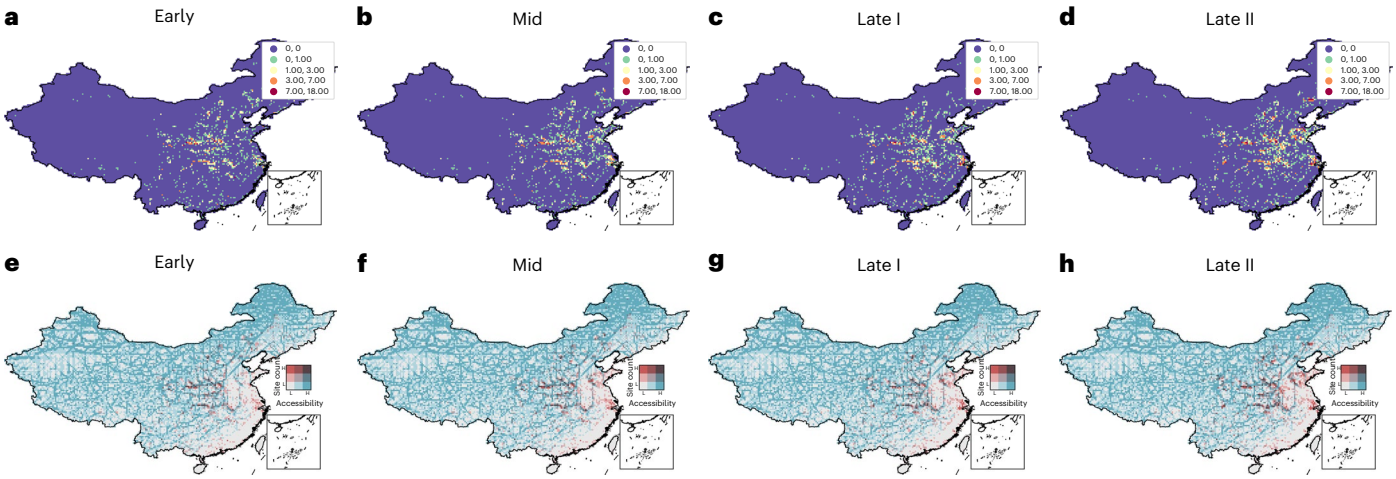



**Fig. 3 | Spatiotemporal evolution of recorded Neolithic settlement intensity and its spatial correspondence with terrain accessibility.** **a**–**d**, Spatial distribution of recorded settlement intensity across the four examined chronological phases: Early Neolithic (**a**), Mid Neolithic (**b**), Late Neolithic I (**c**) and Late Neolithic II (**d**). The intensity of the color gradient represents the kernel density of recorded sites within each period. **e**–**h**, Bivariate choropleth maps illustrating the spatial correspondence between the terrain-accessibility index and recorded settlement intensity for the Early Neolithic (**e**), Mid Neolithic (**f**), Late Neolithic I (**g**) and Late Neolithic II (**h**). Basemaps cited from Standard Map Service, Ministry of Natural Resources of the People's Republic of China (no. GS(2024)0650; https://bzdt.ch.mnr.gov.cn/).

**Table 1 | Terrain accessibility and recorded Neolithic settlement intensity: panel and period-specific regression estimates**

| Period | (1) Pooled (5000–2000 BCE) | (2) Early Neolithic (5000–4000 BCE) | (3) Mid Neolithic (4000–3000 BCE) | (4) Late Neolithic I (3000–2500 BCE) | (5) Late Neolithic II (2500–2000 BCE) |
|---|---|---|---|---|---|
| **Panel (a) (All China)** | | | | | |
| Terrain-accessibility index | **0.136**** | **0.133**** | **0.167***** | **0.163***** | **0.083** |
| Clustered standard error | *(0.056)* | *(0.054)* | *(0.061)* | *(0.056)* | *(0.069)* |
| Two-sided *P* value | 0.014 | 0.013 | 0.006 | 0.004 | 0.232 |
| Observations | 63,236 | 15,809 | 15,809 | 15,809 | 15,809 |
| **Panel (b) (core agricultural region)** | | | | | |
| Terrain-accessibility index | **0.426**** | **0.413**** | **0.470**** | **0.462**** | **0.358** |
| Clustered standard error | *(0.183)* | *(0.173)* | *(0.195)* | *(0.181)* | *(0.229)* |
| Two-sided *P* value | 0.020 | 0.017 | 0.016 | 0.011 | 0.119 |
| Observations | 24,220 | 6,055 | 6,055 | 6,055 | 6,055 |
| **Model specification** | | | | | |
| Provincial fixed effects | Yes | Yes | Yes | Yes | Yes |
| Period fixed effects | Yes | No | No | No | No |
| Geographic controls | Yes | Yes | Yes | Yes | Yes |

Dependent variable: recorded site count per 0.25°×0.25° grid cell. OLS estimates of recorded site counts per 0.25°×0.25° grid cell on the normalized terrain-accessibility index. Geographic controls are longitude, latitude, elevation, slope, caloric productivity and coastal proximity. All models include provincial fixed effects; only the pooled models include period fixed effects. Standard errors clustered at the 2° grid level are in the parentheses. *P* values are unadjusted, two-sided cluster-robust Wald-test *P* values. Observations denote grid-cell-period observations in the pooled models and grid-cell observations in the period-specific models. ***$P < 0.01$; **$P < 0.05$; *$P < 0.1$. Boldface denotes coefficient estimates; italics denote clustered standard errors.

First, global Moran's *I* indicates significant positive spatial autocorrelation across all periods (one-sided permutation $P = 0.001$), steadily rising from 0.2384 in the Early Neolithic to 0.3340 in Late Neolithic II (Table 2(a)). This confirms that the recorded macroscopic settlement distribution became increasingly clustered over time. LISA diagnostics further decompose this trend, revealing a temporal shift from more dispersed local configurations in earlier phases to more concentrated high–high (HH) clusters within localized core regions (Fig. 5).

However, the accessibility-stratified point patterns reveal a non-monotonic shift underlying the linear regression models (Fig. 4b,c and Table 2(b)). During the Early and Mid Neolithic, the highest-accessibility group (Q4) consistently contained the largest number of sites and exhibited the shortest mean nearest-neighbor distance, indicating the strongest local clustering.

This monotonicity began to dissolve in Late Neolithic I and inverted by Late Neolithic II. By Late Neolithic II, the middle-to-high accessibility strata (Q2 and Q3) encompassed the vast majority of sites, with Q2 recording the shortest nearest-neighbor distance (26.9 km). Q4, although still densely populated, was no longer the primary locus of concentration. The LISA maps visualize the accompanying spatial polarization: Late Neolithic I contains localized HH hotspots with relatively few cold spots (Fig. 5c), whereas Late Neolithic II shows the widespread emergence of low–low (LL) clusters (Fig. 5d). As settlement became concentrated in a small number of dense regional cores,

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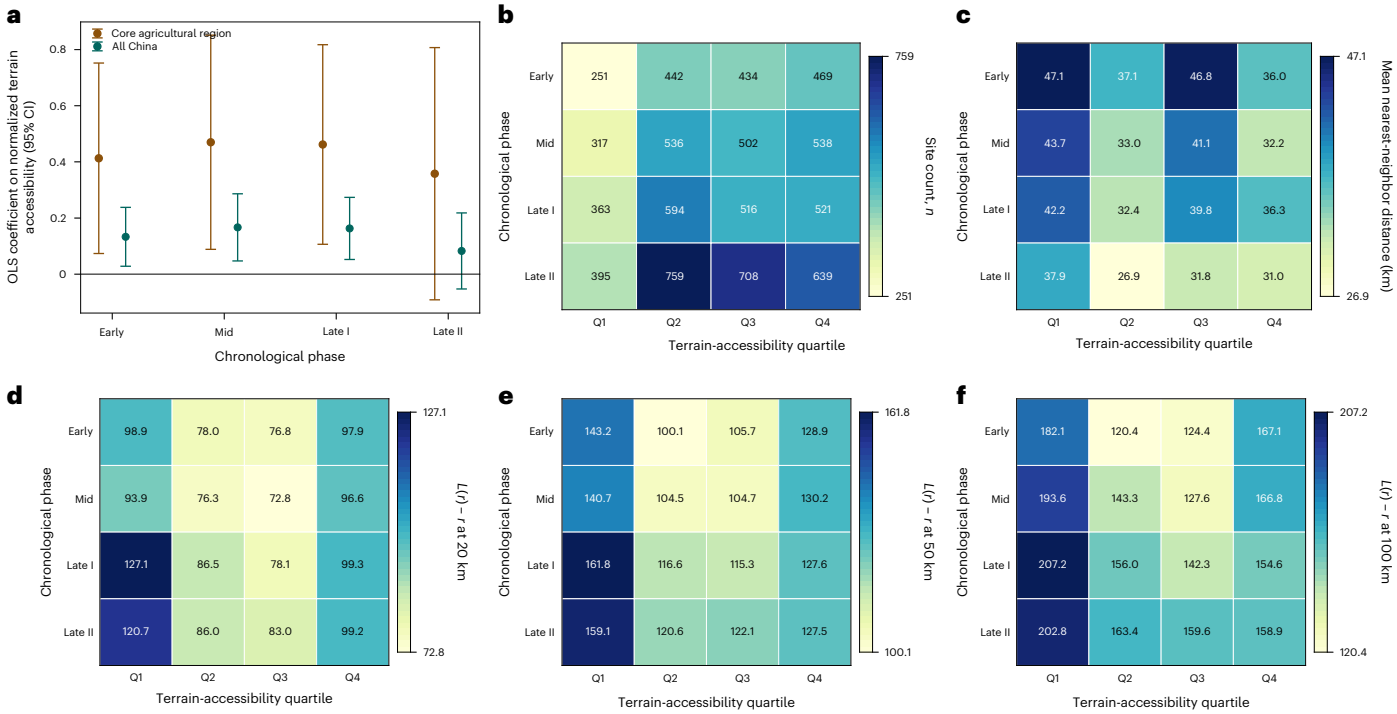



**Fig. 4 | Spatial regression estimates and multiscalar clustering dynamics of recorded Neolithic settlements. a**, Phase-specific OLS estimates of the association between normalized terrain accessibility and recorded site counts per 0.25° × 0.25° grid cell. Points show the estimated coefficient on terrain accessibility; error bars show two-sided 95% confidence intervals (CI) calculated as estimate ± 1.96 × the standard error clustered at the 2° grid level. Each model includes longitude, latitude, elevation, slope, caloric productivity, coastal proximity and province fixed effects. Sample sizes are $n$ = 15,809 grid cells for each phase in All China and $n$ = 6,055 grid cells for each phase in the core agricultural region. The horizontal line indicates a coefficient of zero. Exact coefficient estimates, clustered standard errors and two-sided $P$ values are reported in Table 1; full model specifications are provided in the "Empirical strategy and model specification" section. **b**–**f**, Heat-map visualizations of spatial point-pattern metrics across four chronological phases (Early to Late II Neolithic, $y$ axis) and quartiles of the normalized terrain-accessibility index (Q1–Q4, $x$ axis). **b**, Site counts, $n$. **c**, Mean nearest-neighbor distance in kilometers. **d**–**f**, Multiscalar clustering intensity quantified by the transformed Ripley's $L$ function, $L(r) - r$ (km). Values denote the magnitude of spatial aggregation evaluated at distinct radii: 20 km (**d**), 50 km (**e**) and 100 km (**f**).

the increase in global mean density rendered extensive low-density peripheries statistically significant as LL cold spots.

Concurrently, the spatial clustering gradient across the accessibility strata flattened. The difference in the mean nearest-neighbor distance between the lowest-accessibility (Q1) and highest-accessibility (Q4) groups narrowed from approximately 11 km in the Early and Mid Neolithic to less than 7 km in Late Neolithic II (Fig. 4c and Table 2(b)). This flattening suggests that although dense HH cores dominated the statistical mean, the broader landscape became more strongly differentiated between dense core areas and extensive low-density surroundings.

Furthermore, multiscale Ripley's $L$ analyses highlight the persistent role of localized, topographically enclosed clusters in low-accessibility regions (Fig. 4d–f and Table 2(b)). At scales of 20–100 km, the lowest-accessibility group (Q1) consistently exhibited high $L$ values, reflecting concentrated settlement clusters in constrained settings such as closed basins or river valleys. By Late Neolithic II, as the linear clustering advantage of high-accessibility corridors (Q4) receded, this nonlinear, multicentric pattern gained greater prominence.

Mathematically, this transition from a linear, monotonic increase to an inverted-U distribution—where peak density shifted to intermediate-accessibility zones (Q2/Q3) alongside strong pocket clusters in Q1—would tend to flatten the slope of a linear ordinary least squares (OLS) specification. The LISA-derived evidence of stronger spatial polarization offers one statistical mechanism consistent with the weaker linear marginal effect observed in Late Neolithic II: the increasing contrast between dense HH cores and extensive LL cold spots reduces the fit of a simple monotonic linear gradient across accessibility strata.

Supplementary diagnostics restricted to the core agricultural region support a similar broad interpretive conclusion: settlement clustering remained pronounced into Late Neolithic II, suggesting that the weaker accessibility coefficient reflects a reorganization of spatial structure rather than a disappearance of clustering, although the exact phase-by-phase LISA configuration is not identical to that of the full sample map (Supplementary Fig. 1 and Supplementary Tables 4.5 and 4.6).

## Discussion

These results, based on recorded site counts, trace a relationship between terrain accessibility and settlement intensity that was not a stable feature of the Neolithic. The Late Neolithic II coefficient marks a weakening of this count-based association—not, by itself, evidence of declining population, reduced activity or weaker social organization. The pattern is more naturally understood as historically contingent: the positive association strengthened from the Early Neolithic to Late Neolithic I, weakened markedly in Late Neolithic II, and the spatial diagnostics identify a reorganization rather than a disappearance of clustering. In the full-sample spatial diagnostics, this reorganization involved stronger spatial polarization, renewed concentration in intermediate-accessibility landscapes and persistent isolated "pocket" clusters in low-accessibility settings. Supplementary regional splits further indicate that the pattern was not spatially uniform and was most clearly expressed in the northern core of the core agricultural region (Supplementary Note 4).

This attenuation is not a mechanical artifact of equal-weight counts. Site-area-weighted specifications (Supplementary Note 2) preserve the same phase-specific pattern: within the limits of unevenly

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**Table 2 | Terrain accessibility and recorded Neolithic settlement clustering: spatial diagnostics by accessibility stratum**

| Panel (a) (global grid-cell spatial autocorrelation) | | | | |
|---|---|---|---|---|
| **Period** | **Analysis unit** | ***n*** | **Global Moran's *I*** | **One-sided permutation *P*** |
| Early Neolithic (5000–4000 BCE) | 0.25°×0.25° grid cell | 16,121 | 0.2384 | 0.001 |
| Mid Neolithic (4000–3000 BCE) | 0.25°×0.25° grid cell | 16,121 | 0.2668 | 0.001 |
| Late Neolithic I (3000–2500 BCE) | 0.25°×0.25° grid cell | 16,121 | 0.2877 | 0.001 |
| Late Neolithic II (2500–2000 BCE) | 0.25°×0.25° grid cell | 16,121 | 0.3340 | 0.001 |

| Panel (b) (site-level point-pattern summaries by terrain-accessibility stratum) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Period** | **Accessibility stratum** | **Site*n*** | **Mean NN distance (km)** | **Clark–Evans *R*** | ***Z*** | **Ripley's *L*(*r*)−*r*(km)** | | | |
| | | | | | | **20 km** | **50 km** | **100 km** | **200 km** |
| Early Neolithic (5000–4000 BCE) | Q1 | 251 | 47.109 | 0.453 | −16.565 | 98.851 | 143.235 | 182.148 | 213.716 |
| | Q2 | 442 | 37.098 | 0.474 | −21.162 | 78.004 | 100.097 | 120.431 | 169.276 |
| | Q3 | 434 | 46.845 | 0.593 | −16.224 | 76.831 | 105.712 | 124.411 | 175.610 |
| | Q4 | 469 | 35.965 | 0.473 | −21.825 | 97.872 | 128.946 | 167.058 | 228.197 |
| Mid Neolithic (4000–3000 BCE) | Q1 | 317 | 43.679 | 0.468 | −18.135 | 93.877 | 140.736 | 193.631 | 237.393 |
| | Q2 | 536 | 33.009 | 0.459 | −23.939 | 76.270 | 104.534 | 143.283 | 196.122 |
| | Q3 | 502 | 41.123 | 0.554 | −19.118 | 72.785 | 104.683 | 127.584 | 184.286 |
| | Q4 | 538 | 32.158 | 0.448 | −24.473 | 96.636 | 130.246 | 166.802 | 222.681 |
| Late Neolithic I (3000–2500 BCE) | Q1 | 363 | 42.208 | 0.482 | −18.870 | 127.145 | 161.805 | 207.241 | 260.787 |
| | Q2 | 594 | 32.445 | 0.474 | −24.513 | 86.473 | 116.581 | 155.990 | 211.538 |
| | Q3 | 516 | 39.813 | 0.542 | −19.886 | 78.108 | 115.328 | 142.298 | 208.721 |
| | Q4 | 521 | 36.312 | 0.497 | −21.960 | 99.333 | 127.573 | 154.581 | 199.151 |
| Late Neolithic II (2500–2000 BCE) | Q1 | 395 | 37.939 | 0.451 | −20.871 | 120.685 | 159.128 | 202.762 | 255.005 |
| | Q2 | 759 | 26.933 | 0.444 | −29.310 | 85.961 | 120.551 | 163.444 | 205.183 |
| | Q3 | 708 | 31.828 | 0.507 | −25.114 | 82.974 | 122.119 | 159.638 | 230.081 |
| | Q4 | 639 | 31.033 | 0.469 | −25.664 | 99.246 | 127.451 | 158.917 | 205.425 |

Panel (**a**) reports global Moran's *I* for recorded site counts across *n* = 16,121 grid cells (0.25° × 0.25°) in each period. Spatial weights are row-standardized Queen-contiguity weights. The reported values are one-sided permutation pseudo-*P* values for the alternative of positive spatial autocorrelation, calculated using 999 random permutations; *P* = 0.001 is the minimum attainable value under this procedure. Panel (**b**) reports the point-pattern summaries for archeological site records with valid chronology, coordinates and assignment to the terrain-accessibility index. Each site record is one observational unit; these are not biological or technical replicates, and the study contains no experimental treatment or control group. Within each period, sites are assigned to four common absolute strata of the normalized terrain-accessibility index: Q1 is the lowest-accessibility stratum and Q4, the highest. Site *n* is the exact number of records in each period–stratum group. Mean NN distance is the observed mean nearest-neighbor distance. Clark–Evans *R* compares the observed mean nearest-neighbor distance with that expected under complete spatial randomness; *R* < 1 and negative *Z* indicate aggregation. The reported *Z* statistics correspond to one-sided lower-tail tests against complete spatial randomness; all have *P* < 0.001. Ripley's *L*(*r*)−*r* is a descriptive measure, reported in kilometers at radii of 20, 50, 100 and 200 km; positive values indicate aggregation at the stated spatial scale. No multiple-comparison adjustment was applied. Sample sizes in panel (**b**) differ from the totals reported in Fig. 2 because Fig. 2 summarizes phase-specific site counts within the 15,810-cell grid sample, whereas panel (**b**) uses individual site records eligible for point-pattern analysis. The reported *n* values, therefore, refer to different analytical units and are not expected to coincide.

reported area data, the Late Neolithic II accessibility coefficient weakens under both count- and area-based specifications, even when larger settlements contribute proportionally more to the dependent variable. The phenomenon is, therefore, robust to the choice of measure, although both measures share the underlying limitations of recorded archeological data.

Read together, the spatial diagnostics clarify the form of this reorganization. In the full-sample diagnostics, settlement aggregation was redistributed across the accessibility gradient rather than undergoing a uniform contraction, producing stronger spatial polarization, renewed concentration in intermediate-accessibility settings (Q2–Q3) and persistent microclusters in topographically constrained, low-accessibility landscapes (Q1). Global Moran's *I* rises from 0.24 in the Early Neolithic to 0.33 in Late Neolithic II, whereas the strict monotonic clustering advantage of the highest-accessibility stratum that characterized the Early and Mid Neolithic dissolves by Late Neolithic II.

For urban studies, these findings show how settlement systems were organized before institutionalized cities emerged, without implying that individual sites were themselves urban. Settlement clustering did not simply intensify along the most accessible terrain corridors. It remained strong in Late Neolithic II, but its main concentration shifted toward landscapes of intermediate accessibility. Terrain-derived accessibility, therefore, shaped proto-urban settlement organization without fixing the geography of agglomeration or determining where cities would later form. Its influence changed as settlement systems evolved.

The Circum–Songshan region provides an attribute-rich regional reference for one possible dimension of this reorganization, although it should not be read as a one-to-one validation of the national model. High-resolution studies of more than 1,900 prehistoric settlements in this region document a long-term process of increasing settlement-size differentiation and spatial reorganization from the Peiligang (7000–5000 BCE), through the Yangshao (5000–3000 BCE) and Longshan (3000–2000 BCE), to the Xia–Shang (2000–1046 BCE)[14–18] periods. The maximum area of top-tier centers increased from approximately 40 ha in the Peiligang period to around 100 ha in the Longshan period and to

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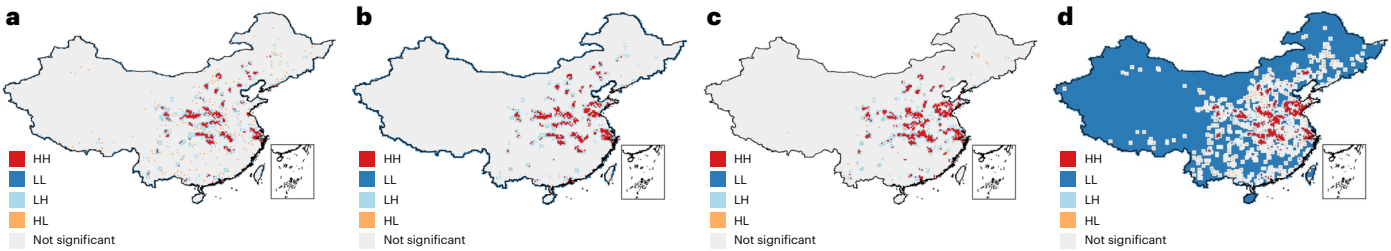



**Fig. 5 | Spatiotemporal evolution of recorded Neolithic settlement clusters in China. a–d**, Local Moran's *I* cluster maps for Early Neolithic (**a**), Mid Neolithic (**b**), Late Neolithic I (**c**) and Late Neolithic II (**d**). The analysis unit is one 0.25° × 0.25° grid cell, with $n = 16{,}121$ grid cells in each phase. Local Moran's *I* was calculated using row-standardized Queen-contiguity spatial weights and 999 conditional random permutations; cells with a one-sided permutation pseudo-$P < 0.05$ are classified as significant. No multiple-comparison adjustment was applied. Red denotes HH clusters, dark blue as LL clusters, light blue as LH spatial outliers and orange as HL spatial outliers; light gray denotes non-significant cells. Basemaps cited from Standard Map Service, Ministry of Natural Resources of the People's Republic of China (no. GS(2024)0650; https://bzdt.ch.mnr.gov.cn/).

as much as 2,500 ha in the Xia–Shang period, whereas the proportion of top-tier settlements among all recorded sites declined.

Settlement nucleation and spatial reorganization are, therefore, one plausible interpretation suggested by the Circum–Songshan evidence: population and activity may have become concentrated in fewer, larger and more differentiated centers. Two observations, however, are not fully accounted for by nucleation alone, indicating that further mechanisms were probably also involved. First, attenuation persists under area-weighted specifications. If a small number of centers had absorbed most late-phase activity and remained within the original high-accessibility corridors, area-weighted measures should at least partly recover the earlier accessibility gradient; instead, the late-phase weakening appears under both count- and area-based specifications. Second, if nucleation had concentrated activity into a few large centers within the original high-accessibility corridors, the point-pattern diagnostics should show intensified clustering within the high-accessibility Q4 stratum. Instead, the strongest local concentration shifted toward intermediate-accessibility settings (Q2–Q3), whereas microclusters persisted in low-accessibility Q1. Nucleation is, therefore, best understood as one of several interacting processes rather than as a sufficient explanation on its own.

A second interpretation concerns regional differentiation. Different regions did not share the same resource bases, geomorphological settings or settlement traditions. The supplementary regional splits (Supplementary Note 4) support this directly: the late-phase weakening is most clearly expressed in the core agricultural region—and especially in its northern core—whereas the southern sample provides much weaker statistical support and does not reproduce the same stable positive gradient. The estimated late-phase weakening was, therefore, spatially uneven rather than evidence of a synchronous national process. Its underlying mechanisms probably differed across the Yellow River systems, Yangtze wetland environments, southeastern coastal zones and northern dryland regions; the national-scale analysis identifies a broad spatial regularity, not a single unified trajectory[6,19].

A third interpretation concerns changes in mobility and land-use practices. The sustained strengthening of settlement concentration along terrain-defined accessible corridors during the earlier phases may reflect the practical advantages of these landscapes for movement and landscape use. By the later phase, however, settlement organization may have placed greater weight on node control, access to hinterlands, risk buffering or defensible locations, thereby weakening the earlier correspondence between recorded settlement intensity and the most accessible corridors[20–23]. Because the index measures a topographic opportunity structure rather than observed movement, these mobility-based readings remain implications of the spatial pattern rather than direct empirical results.

Environmental conditions may also have influenced the timing or intensity of the transition. Hydroclimatic fluctuations around 4.2-ka BP broadly overlap with the terminal Neolithic and are commonly associated with aridity, cooling and heightened regional instability[10,11]. A growing body of research, however, emphasizes that the timing, magnitude and social consequences of this episode varied substantially across regions and cannot be reduced to a simple "global megadrought"[12,13,24–26]. The temporal coincidence between the late-phase weakening and the 4.2-ka interval is consistent with environmental change playing a contextual role, although a direct test lies beyond the present design.

Other interpretations—changes in resource distribution, ritual practice or shifts in shared belief systems—may also have contributed to the observed pattern, but the present site-count data cannot evaluate these mechanisms independently.

These interpretations are not mutually exclusive, and the present design does not allow a definitive adjudication among them. They differ, however, in the directness with which the available evidence can address them. The full-sample spatial diagnostics directly document persistent clustering, a shift toward intermediate accessibility and stronger polarization; the regional splits establish that the accessibility–settlement relationship was spatially uneven; and the area-weighted specifications show that the late attenuation was not solely an artifact of equal-weight site counts. Settlement nucleation, changing mobility and land use, environmental conditions, and archeological visibility may all have contributed, but distinguishing among them will require more consistent evidence on site size and function, material-culture distributions, survey coverage and local environmental change. The central implication is, therefore, not a linear decline caused by a single mechanism, but a historically contingent reorganization of the relationship between landscape accessibility and the proto-urban settlement systems that formed part of the longer prehistory of urban development.

## Conclusion

Taken together, the results indicate that terrain accessibility and settlement organization in Neolithic China were closely related, but not in a temporally fixed way. From the Early Neolithic through Late Neolithic I, settlement intensity became progressively concentrated in highly accessible landscapes, indicating an increasingly strong association between terrain-defined accessibility and settlement location. By Late Neolithic II, however, this long-standing relationship had weakened markedly. Settlement clustering did not disappear: in the full-sample spatial diagnostics, it was expressed through stronger spatial polarization, a shift in the principal concentration of settlements toward intermediate-accessibility landscapes, and the continued presence of localized clusters in low-accessibility environments. Regional analyses further indicate that this phase-specific pattern was most clearly expressed in the northern core of the core agricultural region, rather than representing a spatially uniform trajectory across Neolithic China.

For urban studies, these findings reveal how wider settlement systems were organized before institutionalized cities emerged, without treating individual archeological sites as urban. Terrain-derived accessibility shaped proto-urban settlement organization, but did not fix the geography of agglomeration or determine where cities would form. In Late Neolithic II, clustering remained strong even as its position within the accessibility gradient changed. A comparable historical contingency is evident after state formation: despite relatively persistent underlying geographical constraints, the spatial distribution of Chinese cities was repeatedly reorganized as political, economic and transport systems changed[27]. Geography, therefore, structured the spatial foundations of urban development without locking them into a permanent form.

Several methodological boundaries qualify this interpretation. Comparability across cells and phases rests primarily on recorded site counts, complemented where source data permit by area-weighted alternatives (Supplementary Note 2); the broad phase-specific pattern is observed under both measures. Site counts alone cannot distinguish among population aggregation, site fragmentation, functional reordering, archeological visibility or survey intensity, whereas area-weighted checks remain constrained by an uneven reporting of site extent in the archeological literature[28]. Robustness across alternative specifications and regional restrictions reduces, but does not eliminate, these uncertainties. The accessibility index is constructed from elevation-derived friction and, therefore, captures terrestrial topographic geometry rather than observed prehistoric movement; maritime mobility and historically variable riverine transport lie outside its direct scope and are addressed only through complementary coastal and river-network controls. The analysis consequently identifies a macro-spatial relationship rather than adjudicating among all possible mechanisms. Future work integrating more consistently reported site-area data, functional and material differentiation, survey coverage information and independent palaeoenvironmental evidence will be better positioned to distinguish among alternative explanations and to trace how proto-urban trajectories varied across regions.

Viewed in a wider comparative frame, these findings also contribute to debates about how early complex societies adjusted to the interplay of environmental pressure, spatial structure and social organization. Research in South Asia, Mesopotamia and Egypt indicates that environmental instability need not produce a single trajectory of societal collapse, but may instead accompany settlement-system reorganization, shifts in regional centers and institutional adjustment at different scales[29–35]. The Chinese case provides a long-run East Asian reference for this wider discussion, although such comparisons remain heuristic and do not imply institutional equivalence. More broadly, the results move terrain accessibility from a general geographical background to a testable dimension of the long-run spatial foundations of cities: geography mattered for proto-urban organization, but the relationship between accessibility and settlement concentration was neither universal nor fixed.

## Methods

### Archeological data selection

In previous quantitative research regarding the long-term social evolution of Neolithic China, scholars have often relied on the Atlas of Chinese Cultural Relics (ACCR) as the primary data source for settlement distribution. As the administrative outcome of the Second National Cultural Relics Census in the 1980s, the ACCR has long served as the standard reference for understanding site distribution in China. However, as research scales have advanced toward spatially explicit mechanisms, the inherent limitations of the ACCR data have become increasingly apparent[12,36]: substantial spatial aggregation bias, uneven survey intensity, vague aggregation to county-level administrative centers and the absence of consistent stratigraphic verification.

To overcome these data limitations, this study constructed a completely new Neolithic settlement dataset, drawing primarily from two categories of high-quality archeological records:

1. Excavated sites. Derived from official excavation reports published over the past 40 years. These sites have undergone systematic scientific excavation with clear stratigraphic sequences and typological dating, with some further supported by radiocarbon determinations. Excavated sites are typically subject to relatively rigorous surface surveys and coring investigations, providing systematic estimates of site area, depositional thickness and diachronic distribution.
2. Sites from systematic regional surveys. "Full-coverage" surveys widely conducted in China in recent years use field methods consistent with modern international standards (grid-based walking surveys at fixed intervals and standardized sherd collection and recording). Representative works include Drennan and Peterson[37], Underhill[8] and Li and Xiang[38].

Through this construction strategy, we have maximally filtered out potential "noise" from early censuses—records that were unverified, vaguely located or inaccurately dated. Compared with the ACCR, although this dataset is smaller in absolute numbers, it possesses a qualitative improvement in validity and precision. Included records are typically georeferenced with higher spatial precision and assigned to cultural phases using stratigraphic, typological, and (where available) radiocarbon evidence.

### Study area and unit of observation

We created a regular grid system at 0.25° × 0.25° resolution spanning the geographical extent of China, harmonized in WGS 84 and yielding 16,121 cells. A quarter degree corresponds to approximately 28 km at the equator. This resolution approximates the order of magnitude of a day's travel in pre-industrial societies, defining each cell as a functional hinterland for resource extraction and local interaction. It is sufficiently granular to capture variations in terrain and agricultural suitability, yet coarse enough to buffer minor geolocation uncertainties inherent in archeological data. Grid cells and archeological site records are observational units; the study does not involve experimental treatments, control groups or experimental replication.

This study distinguishes between "China" (the full modern national territory) and "the core agricultural region," which corresponds roughly to the historical core of agriculturally based settlement consolidated during the Qin and Han dynasties—the Yellow River, Yangtze and Pearl River drainages, excluding steppe, desert and high-plateau zones[39–41]. The core agricultural region serves as the principal analytical focus, representing the heartland in which early agricultural settlements, proto-urban centers and long-distance interaction networks first emerged (Fig. 1b). The remainder is used for contextual comparison and robustness checks.

### Study period

In Chinese archeology, the Neolithic period is typically divided into the pre-Yangshao period (before 5000 BCE), the Yangshao period (5000–3000 BCE) and the Longshan period (3000–2000 BCE). Due to the limited number of reliably dated and excavated sites from the pre-Yangshao phase, we exclude it from the scope of this study.

The Yangshao period (5000–3000 BCE) marks a critical phase in the emergence of complex societies in Neolithic China. On the basis of internal developmental differences, we divide the Yangshao period—and its contemporaneous regional cultures (for example, Daxi and Qujialing)—into two sub-phases: Early Neolithic (5000–4000 BCE; early Yangshao and contemporaries) and Middle Neolithic (4000–3000 BCE; late Yangshao and contemporaries).

The Longshan period (3000–2000 BCE) witnessed more pronounced transformations. We similarly subdivide the Longshan

period—and contemporaneous cultures such as Shijiahe and Liangzhu—into Late Neolithic I (3000–2500 BCE; early Longshan) and Late Neolithic II (2500–2000 BCE; late Longshan).

This four-phase chronological framework is constructed primarily as a heuristic compromise for cross-regional macro-comparative analysis under current data conditions. Boundaries of specific periods remain debated; for instance, the lower limit of the Longshan culture in some regions may extend into the 1900s (BCE). Furthermore, due to the lack of widespread absolute radiocarbon dating, the chronological determination of many sites still relies on ceramic typology and stratigraphic comparison[36]. A finer six-phase robustness specification is reported in Supplementary Note 4. We note that the six-phase scheme cannot be uniformly applied to every site—some sites retain four-phase assignments in which finer dating is unavailable—and we, therefore, treat the six-phase analysis as a robustness check rather than a replacement of the four-phase main specification.

### Construction of the terrain-accessibility index

The terrain-accessibility index is a terrain-derived measure of overland geometric accessibility, computed using the Natural Road Score (NRS) algorithm[2]. It is built from a topographic cost surface for a domain surrounding China (70°–140° E, 15°–55° N) to reduce edge effects from modern national boundaries. Elevation inputs are taken from the FAO Global Agro-Ecological Zones (GAEZ) dataset[42] at 1/12° resolution (~10 km at the equator). Because our analysis targets cross-regional terrain-corridor structure rather than fine-grained pedestrian route choice, we implement the index at this coarser resolution to emphasize persistent, first-order geomorphic constraints (for example, valleys and passes) consistent with the macro-spatial observational grid.

Travel time across the terrain is estimated using Langmuir's walking-time function[43]: a base horizontal speed of 1 m per 0.72 s, plus 6 s per meter of ascent; descent saves 2 s per meter on gentle slopes (≤21.25%) and adds 2 s per meter on steeper slopes (>21.25%). These assumptions yield a friction surface approximating pre-mechanized terrestrial movement costs.

Using this friction surface, we apply Dijkstra's algorithm[44] to compute least-cost paths between all pairs of points within a 600-h walking-time radius (up to 2,000–3,000 km, depending on the terrain). This radius is chosen on geometric grounds: it is the spatial extent at which macro-topographic corridor structure is reliably recovered as betweenness-central nodes within the resulting path network. Shorter radii are dominated by local terrain noise and recover only fragmentary, sub-regional structure. The radius is a property of the geomorphic scale at which the index is defined, not a claim about realized travel ranges in any historical period. The index value of a grid cell is the betweenness centrality of the resulting natural path network, normalized to the [0, 1] interval. By construction, the index depends only on terrain inputs and is predetermined with respect to observed settlement locations.

The index is defined over the terrestrial topographic surface only. Incorporating waterborne paths would require non-topographic assumptions—relative water-versus-land travel costs, segment-specific navigability, and seasonal and directional asymmetries—that lie outside the elevation data on which the index is constructed. Rather than fold these heterogeneous assumptions into a single composite measure, we keep the index as a strictly terrain-derived quantity and treat waterborne effects through separate, complementary controls: a coastal proximity dummy in all baseline regressions, and a historical river-network sensitivity check (Supplementary Note 5). Short-range waterborne activity in the Chinese Neolithic—for example, the early dugout canoes from Kuahuqiao in coastal Zhejiang[45]—is captured through these complementary controls rather than embedded in the index. We also do not impose explicit penalties for river crossings, given the long-run variability of riverine systems over millennia.

In this paper, corridors, accessibility and mobility potential refer to this terrain-defined structure—the geometric property of where macro-topography concentrates least-cost overland paths—not to reconstructed transport, communication or interaction systems.

### Parameters and variables

**Dependent variable: recorded settlement intensity.** The number of recorded archeological sites per period within each cell. Because the cell area is fixed, this measure is proportional to recorded site density and captures macroscale patterns of recorded spatial occupation through time. Records of "complete site extent" are highly inconsistent in the source database; chronological attribution can be determined more consistently across cultural phases. We, therefore, use site density as a more robust and comparable metric and report the size-sensitive sensitivity analyses in Supplementary Note 2.

Explanatory variable: terrain-accessibility index (defined above).

Control variables.

- Caloric productivity: maximum potential caloric yield (tons per hectare) from FAO GAEZ for the three most prevalent crops in Neolithic China—wetland rice, foxtail millet and broomcorn millet (incorporating wheat for applicable late phases)—adjusted using FAO caloric values[46].
- Elevation: SRTM 1-arcsecond data[47].
- Slope: mean terrain slope within a grid cell.
- Coastal dummy: binary indicator for cells within 200 km of the coast (robustness to 50-km and 100-km thresholds; Supplementary Note 3).
- Spatial coordinates: longitude and latitude.
- Province-level fixed effects. Their inclusion is pragmatic—provinces partly absorb differences in archeological reporting and research intensity—and does not imply that modern provincial boundaries correspond to prehistoric political units.
- Phase fixed effects are included only in the pooled specification.

### Empirical strategy and model specification

For the pooled model, we estimate

$$N_{i\,t} = \beta_t r_i + X_i'\Theta_t + \lambda_{\mathrm{p}} + \mu_t + \varepsilon_{it}.$$

where $N_{i,t}$ is the recorded settlement intensity in cell $i$ at phase $t$; $r_i$ is the terrain-accessibility index; $X_i$ contains longitude, latitude, elevation, slope, caloric productivity and coastal proximity; $\lambda_{\mathrm{p}}$ denotes the province fixed effects; and $\mu_t$ denotes phase fixed effects. The pooled coefficient is $\beta$.

For each phase $t$, we then separately estimate

$$N_{i,t} = \beta_t r_i + X_i'\Theta_t + \lambda_p + \varepsilon_{i,t}.$$

The phase-specific coefficient $\beta_t$ is plotted in Fig. 4a and reported in Table 1; phase fixed effects are omitted because estimation is within a single phase. Standard errors are clustered at the 2° grid level. The pooled model contains 63,236 complete-case grid-phase observations (24,220 in the core agricultural region); each phase-specific model contains 15,809 observations for All China and 6,055 for the core agricultural region.

### Spatial point-pattern diagnostics

**Clark–Evans nearest-neighbor index.** $R = \bar{r}_{\mathrm{A}}/\bar{r}_{\mathrm{E}}$, where $\bar{r}_{\mathrm{A}}$ is the observed mean nearest-neighbor distance and $\bar{r}_E = 1/(2\sqrt{\rho})$ is the expected mean under a Poisson process with intensity $\rho$. $R < 1$ indicates clustering; significance is evaluated using a one-sided lower-tail $Z$-test against complete spatial randomness.

**Global Moran's *I*.** Standard formulation with a row-standardized Queen-contiguity weights matrix; positive values indicate global spatial clustering. The reported *P* values are one-sided permutation pseudo-*P* values for positive spatial autocorrelation, calculated from 999 random permutations.

**LISA.** Local Moran's $I_i$ identifies four cluster types: HH, LL, high–low (HL) and low–high (LH) ($P < 0.05$).

**Ripley's *K* and *L*.** $L(r) = \sqrt{K(r)/\pi} - r$. Values $L(r) > 0$ indicate clustering at radius *r*. Reported at 20 km, 50 km and 100 km (community, regional and inter-regional scales, respectively) and at 200 km (macro-regional summary). Because the evaluated range is 5–200 km, the reported value at 200 km is the maximum within the inspected range, not necessarily the intrinsic peak.

### Software and computational implementation

Data processing and statistical analyses were conducted in Python (v. 3.12.3) using NumPy (v. 1.26.4), pandas (v. 3.0.3), SciPy (v. 1.17.0), patsy (v. 1.0.2) and statsmodels (v. 0.14.6). Spatial data processing and spatial autocorrelation analyses used GeoPandas (v. 1.1.2), Shapely (v. 2.1.2), Pyogrio (v. 0.12.1), mapclassify (v. 2.10.0), libpysal (v. 4.14.1) and esda (v. 2.8.1). Analytical plots and maps were generated using Matplotlib (v. 3.10.8) and Seaborn (v. 0.13.2). The terrain-accessibility index was constructed following the NRS implementation described above and deposited in Harvard Dataverse.

Generative artificial intelligence tools (Codex and OpenCode, using models from OpenAI, Anthropic and DeepSeek) assisted with data organization and validation, code development, replication-package preparation, consistency checks and language editing. All outputs were reviewed and validated by the authors, who retained full responsibility for the data, analysis and manuscript. Archeological records assembled or checked with artificial intelligence assistance were verified by the authors against the original sources.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

## Data availability

The data supporting this study are available via Zenodo at https://doi.org/10.5281/zenodo.21552061 (ref. 48). The terrain-accessibility measure follows the NRS implementation deposited in the Harvard Dataverse (https://doi.org/10.7910/DVN/P81WBZ); environmental and agricultural data were obtained from FAO GAEZ (https://gaez.fao.org/); elevation data came from USGS SRTM (https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-shuttle-radar-topography-mission-srtm); and historical river-network data came from China Historical GIS Version 6 (https://chgis.fas.harvard.edu/data/chgis/v6/). The underlying record-level archeological database cannot be redistributed because it contains precise site locations and third-party records subject to legal, licensing and heritage-protection restrictions, whereas the aggregated 0.25° grid-level data, spatial weights and machine-readable source data are publicly available in the repository. The base-map boundary data used in Figs. 1, 3 and 5, as well as Supplementary Fig. 1, were obtained from the Standard Map Service of the Ministry of Natural Resources of the People's Republic of China, provided by the National Geomatics Center of China (map approval number GS(2024)0650; https://bzdt.ch.mnr.gov.cn/.

## Code availability

The code supporting the analyses and figure generation reported in this study is publicly available through Zenodo at https://doi.org/10.5281/zenodo.21552061 (ref. 48).

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## Author contributions

R.C., X.Y. and D.L. contributed to early discussions related to the research. R.C. subsequently conceived and designed the study, developed the analytical framework, conducted the statistical and spatial analyses, and drafted the original manuscript. X.Y. provided the original archeological site dataset. R.C. and D.L. used this dataset as a reference catalog, systematically reviewed the corresponding published archeological survey and excavation reports; verified, corrected, and standardized the records; and constructed the final analytical dataset used in the study. K.D. and J.R. supervised the research. During the revision process, X.Y. compiled and synthesized the published archeological literature underpinning the Circum–Songshan case study. R.C., X.Y., H.X. and G.J. contributed to the development and evaluation of alternative hypotheses and interpretations considered during the revision. R.C. conducted all additional analyses, substantially rewrote the manuscript and prepared the responses to reviewers and editors.

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## Competing interests

The authors declare no competing interests.

## Additional information

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

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| ☐ | ☒ | A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
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| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☐ | ☒ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ | Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated |

*Our web collection on statistics for biologists contains articles on many of the points above.*

## Software and code

Policy information about availability of computer code

| | |
|---|---|
| Data collection | Python 3.12.3; Microsoft Excel (Microsoft 365). Generative AI tools (Codex and OpenCode, using models from OpenAI, Anthropic and DeepSeek) assisted with candidate-record extraction and consistency checks. All extracted records were manually verified against the original archaeological sources before inclusion. |
| Data analysis | Python 3.12.3 (NumPy 1.26.4, pandas 3.0.3, SciPy 1.17.0, statsmodels 0.14.6, GeoPandas 1.1.2, Matplotlib 3.10.8, etc.; full list with version numbers in Methods). Generative AI tools (Codex and OpenCode, using models from OpenAI, Anthropic and DeepSeek) assisted with code drafting, debugging and data-audit assistance. All reported results were generated and verified from Python outputs. |

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The data supporting this study are available through Zenodo at https://doi.org/10.5281/zenodo.21552061. The terrain-accessibility measure follows the Natural Road Score implementation deposited in Harvard Dataverse (https://doi.org/10.7910/DVN/P81WBZ); environmental and agricultural data were obtained from FAO Global Agro-Ecological Zones (https://gaez.fao.org/); elevation data came from USGS SRTM (https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-shuttle-radar-topography-mission-srtm); and historical river-network data came from China Historical GIS Version 6 (https://chgis.fas.harvard.edu/data/chgis/v6/). The underlying record-level archaeological database cannot be redistributed because it contains precise site locations and third-party records subject to legal, licensing and heritage-protection restrictions, while the aggregated 0.25° grid-level data, spatial weights and machine-readable Source Data are publicly available in the repository.

## Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

| | |
|---|---|
| Reporting on sex and gender | *Use the terms sex (biological attribute) and gender (shaped by social and cultural circumstances) carefully in order to avoid confusing both terms. Indicate if findings apply to only one sex or gender; describe whether sex and gender were considered in study design; whether sex and/or gender was determined based on self-reporting or assigned and methods used.*<br>*Provide in the source data disaggregated sex and gender data, where this information has been collected, and if consent has been obtained for sharing of individual-level data; provide overall numbers in this Reporting Summary. Please state if this information has not been collected.*<br>*Report sex- and gender-based analyses where performed, justify reasons for lack of sex- and gender-based analysis.* |
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| Population characteristics | *Describe the covariate-relevant population characteristics of the human research participants (e.g. age, genotypic information, past and current diagnosis and treatment categories). If you filled out the behavioural & social sciences study design questions and have nothing to add here, write "See above."* |
| Recruitment | *Describe how participants were recruited. Outline any potential self-selection bias or other biases that may be present and how these are likely to impact results.* |
| Ethics oversight | *Identify the organization(s) that approved the study protocol.* |

Note that full information on the approval of the study protocol must also be provided in the manuscript.

# Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☒ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

# Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

| | |
|---|---|
| Study description | This is a quantitative observational study based on a secondary compilation of published archaeological settlement records from Neolithic China. The study examines the relationship between terrain-derived accessibility and settlement organization across archaeological phases between c. 5000 and 2000 BCE. |
| Research sample | The research sample comprises 3,587 georeferenced Neolithic settlement records from China, dated to approximately 5000-2000 BCE. Records were compiled from published archaeological surveys, excavation reports and regional archaeological studies. |

| | |
|---|---|
| Sampling strategy | No random sampling was conducted. |
| Data collection | Xiao Yu provided the original archaeological site dataset and the corresponding reference list of published sources. Ruoran Cheng and Dongdong Li used this dataset as a catalogue, systematically reviewed the corresponding published excavation reports and regional survey publications, manually verified, corrected and standardised each record against the original sources, and constructed the final analytical dataset. Generative AI tools assisted with candidate-record extraction and consistency checks, but all extracted records were manually verified against the original archaeological sources before inclusion (see "Software and computational implementation" in Methods). Each record includes a chronological attribution indicating the phase(s) in which the site was occupied (a single site may be present in multiple phases), with phase assignments based on stratigraphic sequences, ceramic typology and, where available, radiocarbon determinations. |
| Timing | The original archaeological-site catalogue and accompanying reference list were provided in mid-2023 by X.Y. The records were systematically reviewed, verified and standardised against the original published sources from mid-2023 to the end of 2023 by R.C. and D.L. Iterative data cleaning, validation and spatial analysis continued from early 2024 until the analytical dataset was finalised at the end of 2024 by R.C. |
| Data exclusions | No eligible site records were excluded from the primary site-count analysis after compilation. |
| Non-participation | N/A |
| Randomization | N/A |

# Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

## Materials & experimental systems

| n/a | Involved in the study |
|---|---|
| ☐ | ☐ Antibodies |
| ☐ | ☐ Eukaryotic cell lines |
| ☐ | ☐ Palaeontology and archaeology |
| ☐ | ☐ Animals and other organisms |
| ☐ | ☐ Clinical data |
| ☐ | ☐ Dual use research of concern |
| ☐ | ☐ Plants |

## Methods

| n/a | Involved in the study |
|---|---|
| ☐ | ☐ ChIP-seq |
| ☐ | ☐ Flow cytometry |
| ☐ | ☐ MRI-based neuroimaging |

## Antibodies

| | |
|---|---|
| Antibodies used | *Describe all antibodies used in the study; as applicable, provide supplier name, catalog number, clone name, and lot number.* |
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## Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

| | |
|---|---|
| Cell line source(s) | *State the source of each cell line used and the sex of all primary cell lines and cells derived from human participants or vertebrate models.* |
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| Commonly misidentified lines (See ICLAC register) | *Name any commonly misidentified cell lines used in the study and provide a rationale for their use.* |

## Palaeontology and Archaeology

| | |
|---|---|
| Specimen provenance | *Provide provenance information for specimens and describe permits that were obtained for the work (including the name of the* |

| | |
|---|---|
| Specimen provenance | *issuing authority, the date of issue, and any identifying information). Permits should encompass collection and, where applicable, export.* |
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| Laboratory animals | *For laboratory animals, report species, strain and age OR state that the study did not involve laboratory animals.* |
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| Ethics oversight | *Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.* |

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| Outcomes | *Describe how you pre-defined primary and secondary outcome measures and how you assessed these measures.* |

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Policy information about dual use research of concern

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|---|---|---|
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|---|---|---|
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| ☐ | ☐ | Increase transmissibility of a pathogen |
| ☐ | ☐ | Alter the host range of a pathogen |
| ☐ | ☐ | Enable evasion of diagnostic/detection modalities |
| ☐ | ☐ | Enable the weaponization of a biological agent or toxin |
| ☐ | ☐ | Any other potentially harmful combination of experiments and agents |

# Plants

Seed stocks

*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*

Novel plant genotypes

*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*

Authentication

*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*

# ChIP-seq

## Data deposition

☐ Confirm that both raw and final processed data have been deposited in a public database such as GEO.

☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links
*May remain private before publication.*

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Files in database submission

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Genome browser session
(e.g. UCSC)

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## Methodology

Replicates

*Describe the experimental replicates, specifying number, type and replicate agreement.*

Sequencing depth

*Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.*

Antibodies

*Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.*

Peak calling parameters

*Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.*

Data quality

*Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.*

Software

*Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.*

# Flow Cytometry

## Plots

Confirm that:

☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☐ All plots are contour plots with outliers or pseudocolor plots.

☐ A numerical value for number of cells or percentage (with statistics) is provided.

## Methodology

| | |
|---|---|
| Sample preparation | *Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.* |
| Instrument | *Identify the instrument used for data collection, specifying make and model number.* |
| Software | *Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.* |
| Cell population abundance | *Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.* |
| Gating strategy | *Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.* |

☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

# Magnetic resonance imaging

## Experimental design

| | |
|---|---|
| Design type | *Indicate task or resting state; event-related or block design.* |
| Design specifications | *Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.* |
| Behavioral performance measures | *State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).* |

## Acquisition

| | |
|---|---|
| Imaging type(s) | *Specify: functional, structural, diffusion, perfusion.* |
| Field strength | *Specify in Tesla* |
| Sequence & imaging parameters | *Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.* |
| Area of acquisition | *State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.* |

Diffusion MRI ☐ Used ☐ Not used

## Preprocessing

| | |
|---|---|
| Preprocessing software | *Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).* |
| Normalization | *If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.* |
| Normalization template | *Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.* |
| Noise and artifact removal | *Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).* |

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*Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.*

## Statistical modeling & inference

Model type and settings

*Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).*

Effect(s) tested

*Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.*

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference

*Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.*

(See Eklund et al. 2016)

Correction

*Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).*

## Models & analysis

| n/a | Involved in the study |
|---|---|
| ☐ | ☐ Functional and/or effective connectivity |
| ☐ | ☐ Graph analysis |
| ☐ | ☐ Multivariate modeling or predictive analysis |

Functional and/or effective connectivity

*Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).*

Graph analysis

*Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).*

Multivariate modeling and predictive analysis

*Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.*